For my Heart Disease Prediction & Signal Analysis midterm project, I designed and built the entire 16-section experimental workflow from raw data preprocessing all the way through model explainability, feature engineering, and controlled signal verification. I started by cleaning the 10,000-patient dataset—imputing missing numerical values with the median and categorical values with the mode—and then implemented a leak-free preprocessing pipeline where I encoded categorical features and split the raw data before fitting StandardScaler strictly on the training set. When my baseline Random Forest model failed to predict the minority class (0.00 recall on positive cases), I personally drove the investigation across multiple imbalance and optimization techniques: I computed balanced class weights, wrote custom Precision-Recall curve logic to test optimal (0.0600) and balanced (0.5000) decision thresholds, applied SMOTE oversampling, and trained an XGBoost classifier with scale_pos_weight alongside 5-fold cross-validation and GridSearchCV hyperparameter tuning. Rather than guessing why the models stayed around a ~0.50 ROC-AUC, I contributed a full diagnostic and explainability section using Pearson correlations, permutation importance, and SHAP TreeExplainer (bar, beeswarm, and dependence plots) to mathematically prove that the raw synthetic target was generated independently of the 20 patient features (all linear correlations within ±0.02, with Stress Level at just 0.016). To push the feature space further, I engineered degree-2 polynomial interactions across key biomarkers, applied SelectKBest and RFECV feature selection, consolidated redundant lipid and blood pressure columns, and created three domain-specific clinical indices from scratch: a Metabolic Risk Score (BMI * Triglyceride Level * Fasting Blood Sugar), a Cardiovascular Strain Index (Blood Pressure * Age), and an Inflammatory-Vascular Score (CRP Level * Homocysteine Level). Finally, to prove that my machine learning architecture was sound, I designed a custom clinical signal verification experiment weighted by real cardiovascular risk factors (Age, BMI, Blood Pressure, Cholesterol Level, Homocysteine Level, Smoking, and Diabetes), added stratified shuffling, cleaned feature names for LightGBM, and built both a weighted Soft-Voting Ensemble (XGBoost + LightGBM + Random Forest) that achieved a 0.9948 ROC-AUC and a final high-capacity Logistic Regression model that reached a 1.0000 ROC-AUC with 100% accuracy, precision, and recall, wrapping the notebook up with a complete comparative summary table in Section 16.